ADT-Net: A Text-Based Person Search Network Utilizing Adaptive Transform

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Motivation and Contribution

Motivation: Previous methods do not explicitly consider the impact of image illumination changes and background information on retrieval performance. In addition, current cross-modal global matching methods are insensitive to challenging negative samples, mainly because of the small inter-class differences.

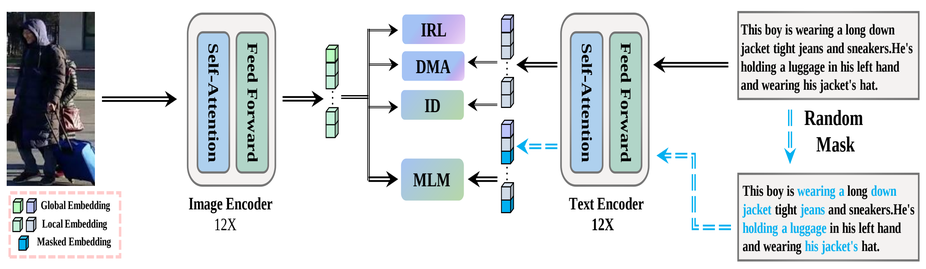


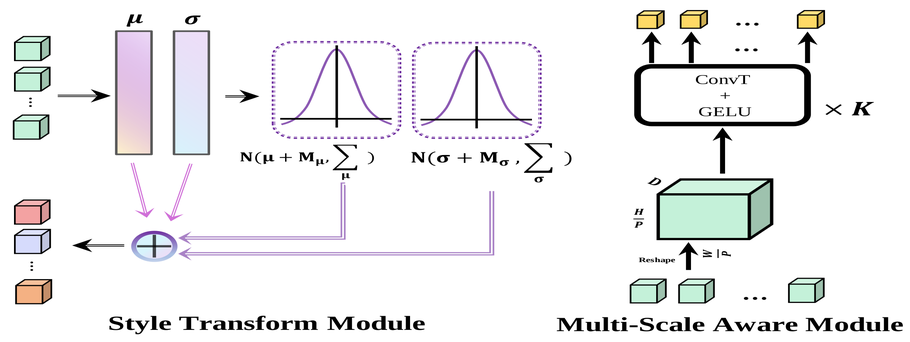
Contributions:

- We design a new module: IRL, which uses the change of style information to simulate the potential illumination changes in the image. Furthermore, we use a multi-scale perception module to reduce the interference of background noise on pedestrian information. This allows the model to better adapt to illumination changes and extract more robust visual features.
- We introduce a dynamic matching alignment module DMA. This module significantly improves the network's ability to discriminate difficult-to-negative samples by applying bidirectional KL divergence and introducing a dynamic weight mechanism.

Our proposed method

Overall Framework of Our Method:





IRL: Invariant Representation Learning Module

To address the impact of image illumination changes and background noise, we designed the IRL module, which consists of two submodules: the ST module and the MSA module.

- ST: The potential illumination variations in pedestrian images are simulated by adjusting the mean and standard deviation of visual features.

$$\hat{V} = \hat{\sigma}(\frac{V-\mu}{\sigma}) + \hat{\mu},$$

where $\hat{\sigma}$ and $\hat{\mu}$ are sampled from Gaussian distributions.

- MSA: Ensure that key pedestrian information is effectively enhanced during the style transfer process, thereby ignoring background noise.

DMA: Dynamic Matching Alignment Module

We define the cross-modal projection matching loss from text to image as:

$$L_{t2i} = KL(p|q) = \sum_{i=1}^{N}\sum_{j=1}^{N} p_{ij} log(\frac{p_{ij}}{q_{ij}})$$

In order to better model the distribution in both directions and thus pay more attention to hard negative samples, we design a bidirectional KL divergence loss with dynamic weights.

$$L_{t2i} = \gamma KL(p|q) + (1-\gamma)KL(q|p),$$

where $= \frac{Cur_Epoch}{Total_Epoch}$. Cur_Epoch is the size of the current Epoch, and Total_Epoch is the size of the total Epoch .

Results on Public Datasets

ICFG-PEDES

Methods	Ref	R@1	R@5	R@10	mAP
Dual Path [30]	TMM20	38.99	59.44	68.41	-
CMPM [27]	ECCV18	43.51	65.44	74.26	-
MIA [31]	TIP20	46.49	67.14	75.18	-
SCAN [32]	ECCV18	50.05	69.65	77.21	-
SSAN [10]	Arxiv21	54.23	72.63	79.53	-
TIPCB [14]	Neuro22	54.96	74.72	81.89	-
IRRA [12]	CVPR23	63.46	80.24	85.82	38.06
CFAM [13]	CVPR24	63.57	80.57	**86.37**	38.34
Our	-	**63.94**	**80.91**	85.77	**40.54**

RSTPReid

Methods	Ref	R@1	R@5	R@10	mAP
DDSL [24]	MM21	39.05	62.60	73.95	-
SSAN [10]	Arxiv21	43.50	67.88	77.15	-
LBUL [15]	MM22	45.55	68.20	77.85	-
IVT [17]	ECCV22	46.70	70.00	78.80	-
CFine [16]	TIP23	50.55	72.50	81.60	-
IRRA [12]	CVPR23	60.20	81.30	88.20	47.17
CFAM [13]	CVPR24	60.51	**82.85**	**89.71**	**47.64**
Our	-	**60.85**	82.60	88.55	47.21

CUHK-PEDES

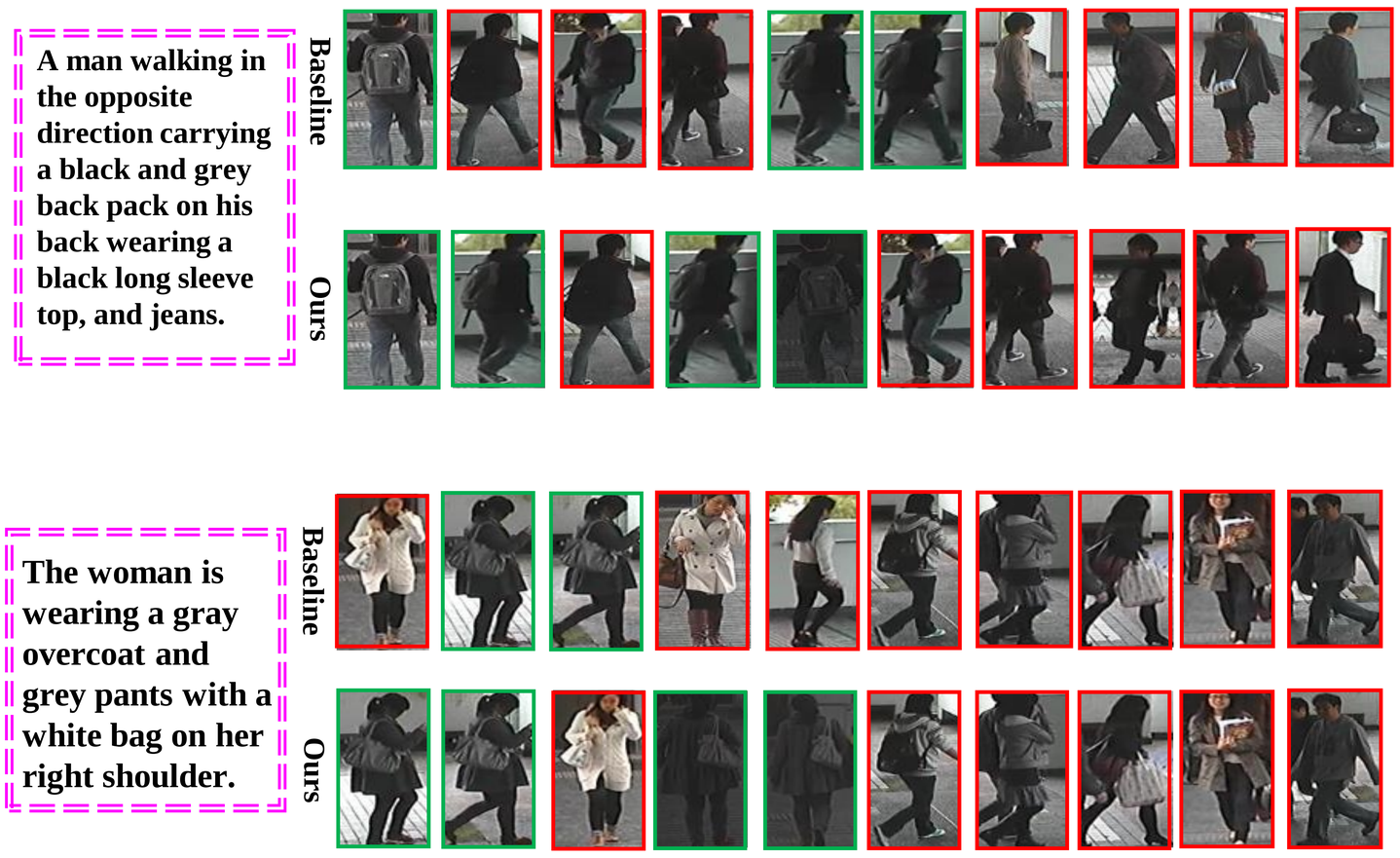
The left table shows the results of R@1, R@5, R@10 and mAP on two public datasets.

Methods	ImageEnc	TextEnc	Ref	CUHK-PEDES				
				R@1	R@5	R@10	mAP	mINP
NAFS [28]	ResNet50	Bi-GRU	Arxiv21	59.94	79.86	86.70	54.07	-
SSAN [10]	ResNet50	LSTM	Arxiv21	61.37	80.15	86.73	-	-
LBUL [15]	ResNet50	BERT	MM22	64.04	82.66	87.22	-	-
TIPCB [14]	ResNet50	BERT	Neuro22	64.26	83.19	89.10	-	-
MANet [11]	ResNet50	BERT	TNNLS23	65.64	83.01	88.78	-	-
ACSA [22]	ResNet50	BERT	TMM22	68.67	85.61	90.66	-	-
CFine [16]	CLIP-ViT-B/16	BERT	TIP23	69.57	85.93	91.15	-	-
IRRA [12]	CLIP-ViT-B/16	CLIP-Xformer	CVPR23	73.38	**89.93**	**93.71**	66.13	50.24
CFAM [13]	CLIP-ViT-B/16	CLIP-Xformer	CVPR24	73.67	89.71	93.57	65.94	51.32
Our	CLIP-ViT-B/16	CLIP-Xformer	-	**73.89**	89.45	93.61	**66.77**	**51.65**

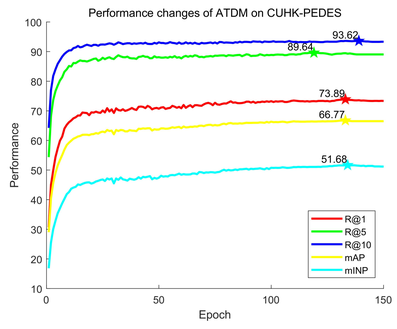
In order to better explore the effectiveness of our method on hard negative samples, we conducted experiments on the CUHK-PEDES dataset and utilized the mINP evaluation metric. As can be seen from the table, our method has a very significant improvement in the mINP index.

Experiments and Conclusion

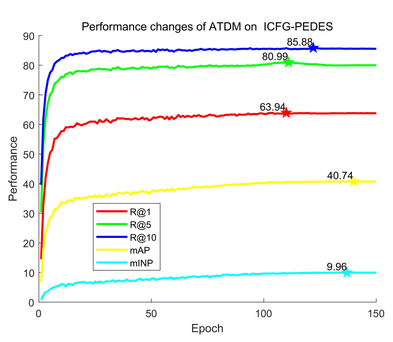
Quantitative Results



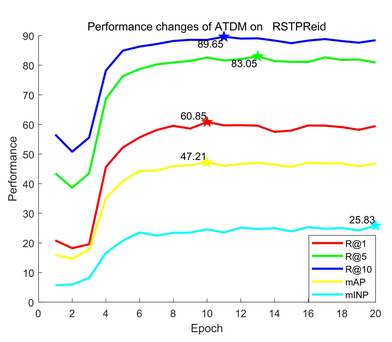
Performance Analysis on CUHK-PEDES



Performance Analysis on ICFG-PEDES



Performance Analysis on RSTPReid



Conclusion

Our approach aims to enhance the network's adaptability to illumination changes in pedestrian images, thereby improving the stability of visual features and addressing the problem that cross-modal global matching is insensitive to hard negative samples. Specifically, we design an IRL module that includes a style transfer (ST) submodule and a multi-scale perception (MSA) submodule. The ST module learns visual features that can adapt to lighting conditions by simulating illumination changes in images. The MSA module enhances the expression of pedestrian features by comparing visual features at different scales and effectively reduces the interference of background noise. In addition, our model also integrates a dynamic bidirectional matching module (DMA) to optimize image-text global feature matching through bidirectional KL divergence and dynamic weight allocation, with special attention to hard negative sample recognition.